Anscombe's Quartet



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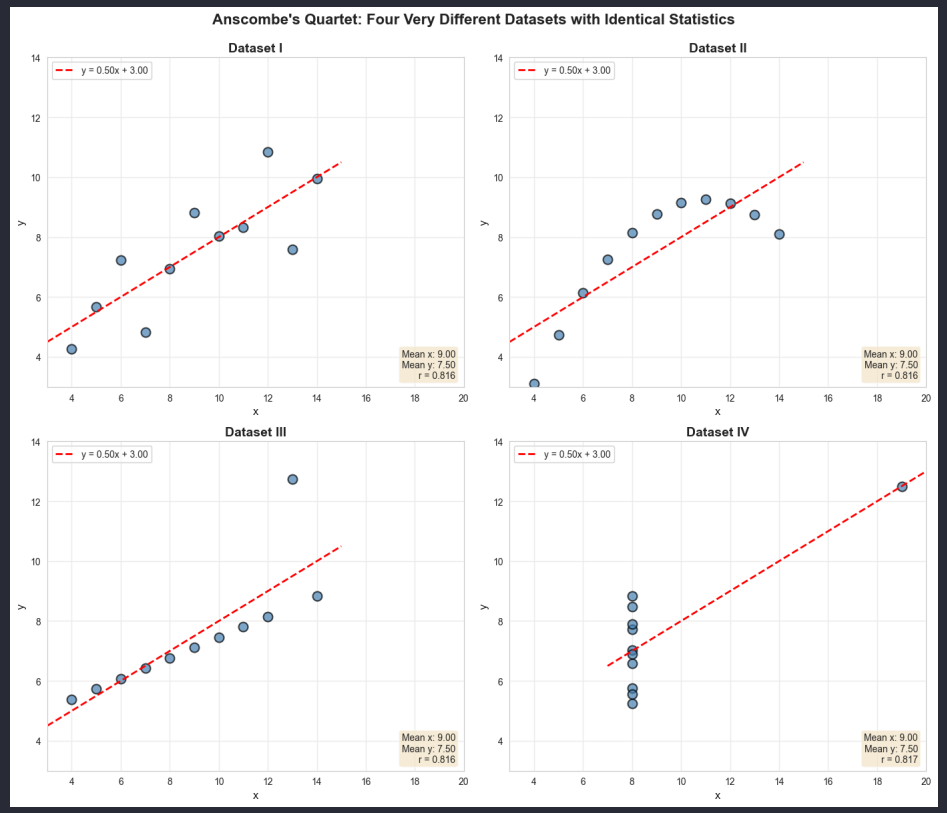
The Setup: Four "Identical" Datasets

- All four datasets have **exactly the same** statistics:
  - Mean of X: **9.0**
  - Mean of Y: **7.5**
  - Standard deviation of X: **3.3**
  - Standard deviation of Y: **2.0**
  - Correlation: **0.816**
  - Linear regression: **y = 0.5x + 3.0**

> ***Question:*** *If the statistics are identical, are the datasets the same?*

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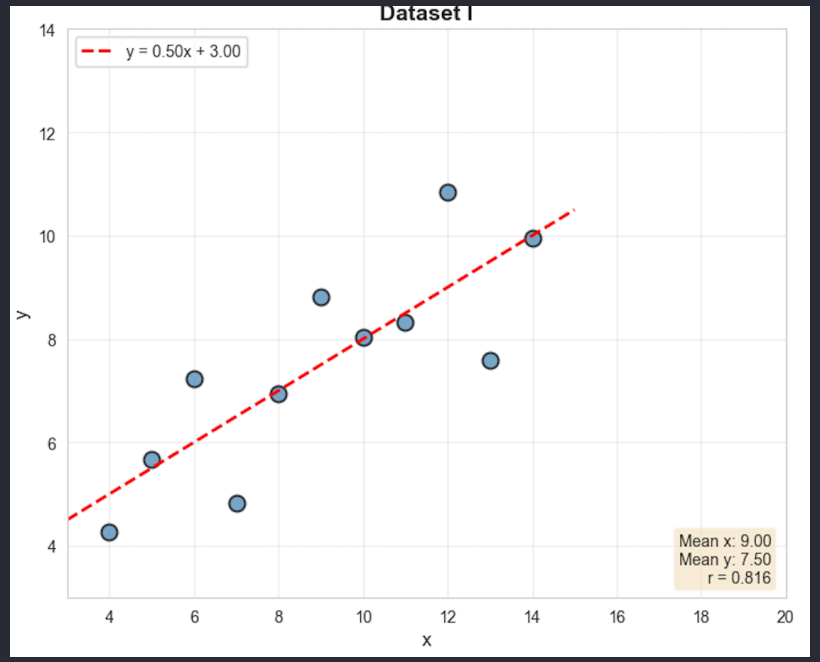
The Reality: Four Completely Different Stories



Always visualize your data!

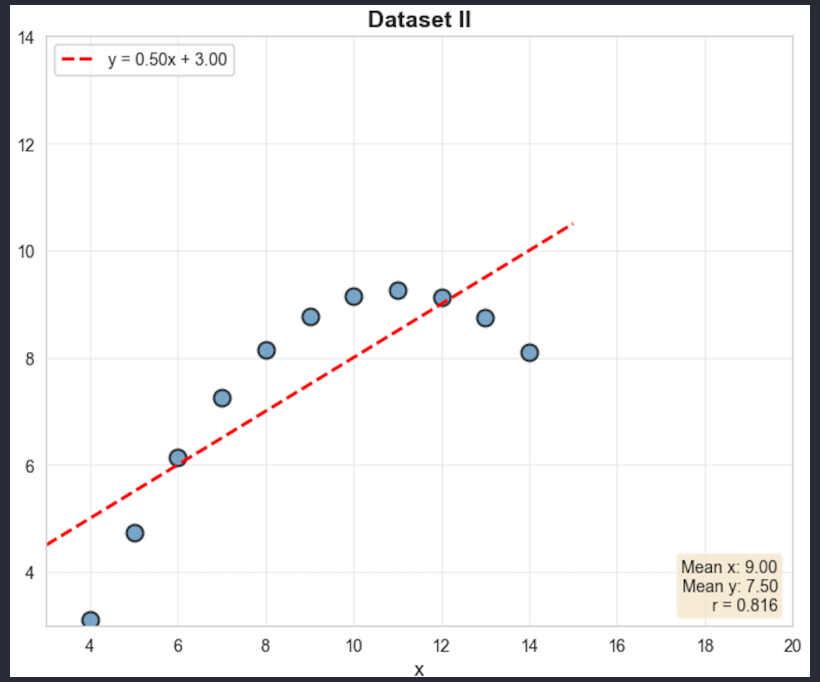
Dataset I: The "Good" One

- **Pattern:** Simple linear relationship with random scatter
- **What it means:**
  - Linear regression is appropriate
  - Predictions will be reasonably accurate
  - This is what we *hope* our data looks like
- **Takeaway:** Even perfect scenarios have variance!



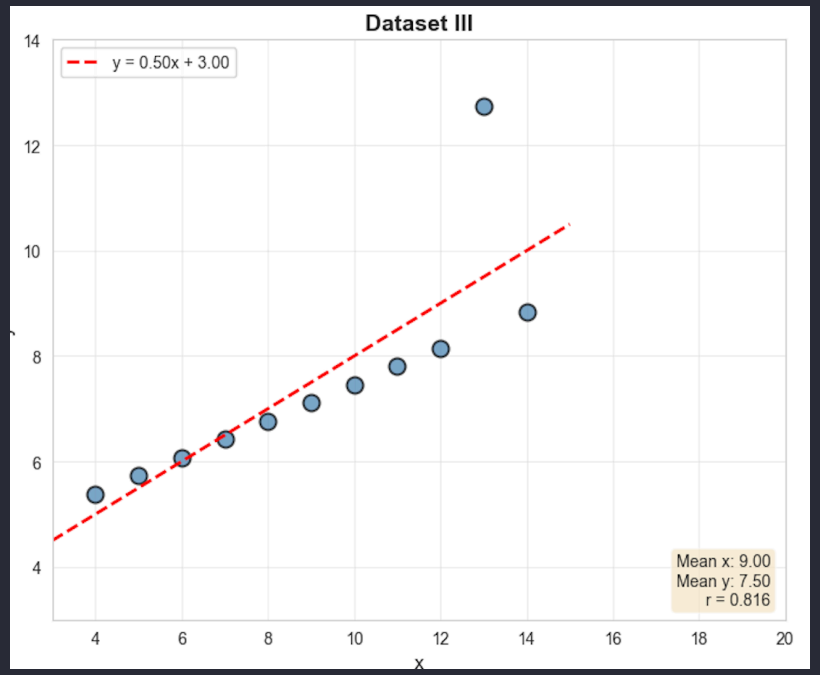
Dataset II: The Curved One

- **Pattern:** Clear non-linear (quadratic) relationship
- **The Problem:**
  - Linear model makes **systematic errors**
  - Underestimates at edges, overestimates in middle
  - U-shaped residual pattern (red flag!)
- **The Fix:**
  - Use polynomial or non-linear regression
  - $R^2 = 0.67$ looks "good" but model is **wrong**



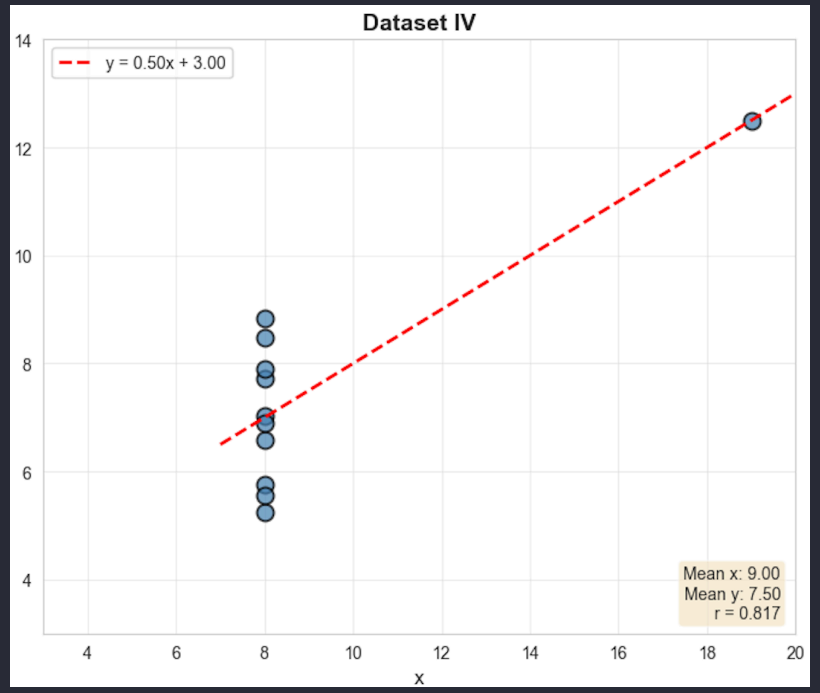
Dataset III: The Outlier

- **Pattern:** Perfect linear... except ONE point
- **The Problem:**
  - Single outlier drastically affects the regression line
  - Rest of data follows a different pattern
  - Outlier has high **leverage** (unusual x-value)
- **Questions to Ask:**
  - Is this a data error?
  - Is this a special case to investigate?
  - Should we use robust regression?



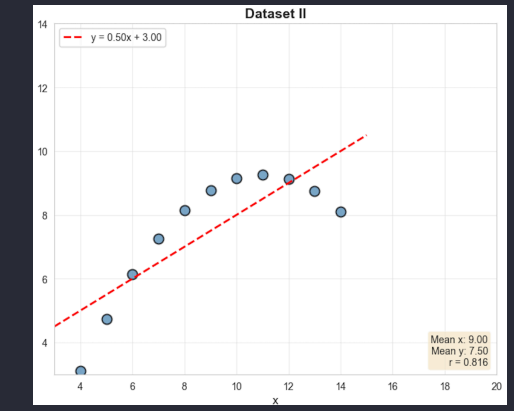
Dataset IV: The Most Dangerous

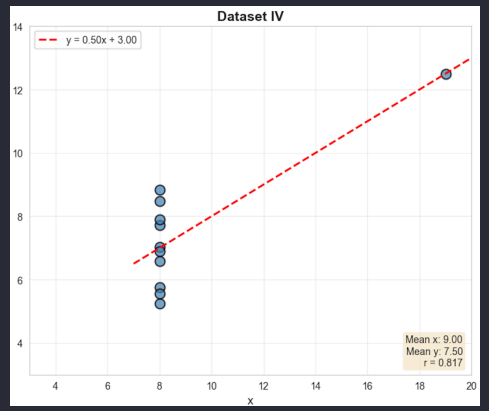
- **Pattern:** NO relationship exists!
- **The Illusion:**
  - 10/11 points have x = 8 (no variance!)
  - ONE outlier at x = 19 creates fake correlation
  - Remove it → correlation drops to ~0
- **Real-World Impact:**
  - Would conclude "X predicts Y" when it doesn't
  - Business decisions based on false relationships
  - Potentially catastrophic outcomes



The Business Scenario

- **Imagine:** Y = Sales, X = Ad Spend
- **Dataset II** (non-linear):
  - Wrong model → wrong budget allocation
  - Miss optimal spending levels
- **Dataset IV** (outlier-driven):
  - Believe ads drive sales when they don't!
  - Waste millions on ineffective campaigns
  - Miss actual drivers of sales
- **Cost:** Potentially millions in poor decisions





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Lesson 1: Summary Statistics Lie

- **Same numbers can tell completely different stories**
  - Mean, std dev, correlation can be identical
  - Yet underlying data is fundamentally different
  - Numbers without context are dangerous
- **Golden Rule:** Never trust statistics alone

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Lesson 2: Visualization is Non-Negotiable

- **Visualization reveals what statistics hide:**
  - Non-linear relationships
  - Outliers and anomalies
  - Clusters and groupings
  - Data quality issues
  - Violations of assumptions
- **Best Practice:** Plot first, model second

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Lesson 3: Check Your Assumptions

- Linear regression assumes:
  1. **Linearity** - relationship is actually linear
  2. **Independence** - observations are independent
  3. **Homoscedasticity** - constant variance
  4. **Normality** - residuals are normally distributed
  5. **No outliers** - or at least account for them
- **If assumptions fail, your predictions fail**

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Lesson 4: Residual Plots Are Your Friend

- **Good residual plot:**
  - Random scatter around zero
  - No patterns or trends
  - Constant spread
- **Bad residual plots reveal:**
  - U-shape → non-linear relationship (Dataset II)
  - Funnel shape → heteroscedasticity
  - Clusters → missing variables
  - Outliers → special cases

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Lesson 5: One Outlier Can Ruin Everything

- **High leverage points** (unusual x-values) have outsized influence:
  - Can create relationships that don't exist
  - Can hide relationships that do exist
  - Can dramatically change model parameters
- **What to do:**
  1. **Investigate** - Why is this point different?
  2. **Don't auto-delete** - It might be your most important finding!
  3. **Consider robust methods** - Less sensitive to outliers
  4. **Report sensitivity** - How much does model change?

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Lesson 6: Good Metrics ≠ Good Model

- **Dataset II:** $R^2 = 0.67$ (looks decent!)
  - But model is **fundamentally wrong**
  - Systematic prediction errors
  - Choosing wrong model type
- **Dataset IV:** $R^2 = 0.67$ (looks decent!)
  - But relationship is **completely false**
  - Driven by single outlier
  - Would make terrible predictions
- **Beware:** Metrics can be misleading without context

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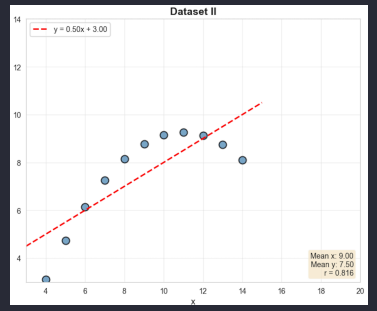
Best Practices

1. Visualize First
2. Explore & Understand
3. State Assumptions
4. Build Model
5. Diagnostic Checks (residuals, outliers)
6. Validate on New Data
7. Maintain Skepticism!

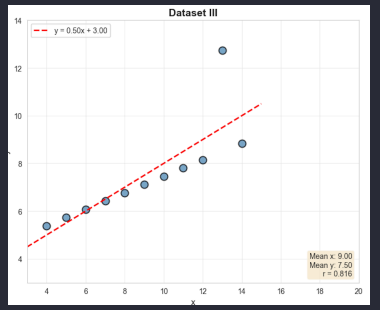
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Real-World Applications

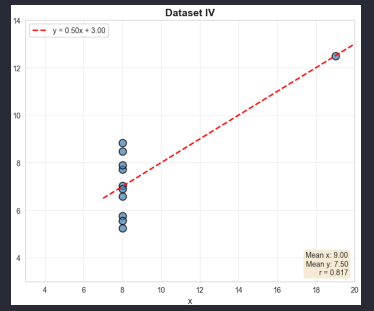
- **Dataset II (Non-linear):**
  - Returns diminishing over scale
  - Learning curves, dose-response
  - Economic relationships



- **Dataset III (Outlier):**
  - Rare events or anomalies
  - Data entry errors
  - Special cases worth studying



- **Dataset IV (False correlation):**
  - Confounding variables
  - Spurious relationships
  - Small sample sizes



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Common Mistakes to Avoid

- Reporting only R^2 without visualizing
- Deleting outliers without investigation
- Assuming linearity without checking
- Ignoring residual patterns
- Using models on data they weren't built for
- Trusting a model just because it has "good" metrics
- Forgetting to validate on new data

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What Makes a Good Data Scientist?

- **Technical Skills:**
  - Know your statistical methods
  - Understand assumptions and limitations
  - Use diagnostic tools effectively
- **Critical Thinking:**
  - Question the data and results
  - Look for what might be wrong
  - Don't fall in love with your models
- **Communication:**
  - Visualize to explain
  - Acknowledge uncertainty
  - Document assumptions

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The Anscombe's Quartet Mindset

> *"The greatest value of a picture is when it forces us to notice what we never expected to see."*
> *— John Tukey*

> *Every dataset is guilty until proven innocent by visualization*

Key Takeaway

- In data science, seeing is believing
- Summary statistics are never enough
- Always **Visualize** your data

Video

- Same Stats, Different Graphs

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References

Topic	Source
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Tukey (1977)	Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley.